

Q: 2(i)

A: 2(i)

Analytical chemistry :-

Analytical chemistry is the branch of chemistry that deals with the methods and ~~tested~~ instruments for determining the composition and ~~structure~~ ^{properties} of matter.

Physical chemistry :-

Is the branch of chemistry that deals with laws and theories to understand the changes and structure of matter.

Q: 2(ii)
A: 2(ii)

homogeneous
mixture

① It has uniform composition throughout.

② It can be only in single phase

③ Components cannot be seen ~~not~~ / distinguished by eye.

④ Example :-

① Salt in water

② Air

heterogeneous
mixture

It has non-uniform composition.

It can be two or more than 2 phases.

Components can be distinguished / seen by naked eye.

Example:-

① Sand in water

② Salad.

Q: 2(iii)
A: 2(iii)

The maximum number of electron a shell can hold is given by the rule $2n^2$.
where n is the shell number.

① K shell = $n=1 \Rightarrow 2n^2 = 2(1)^2$
 $= 2(1)$

2 electrons

② L-shell = $n=2 \Rightarrow 2n^2 = 2(2)^2 \Rightarrow 2(4)$

8 electrons

$$M\text{-shell} = 2n^2 = n=3 \rightarrow 2(3)^2 \Rightarrow 2(9) \\ 18 \text{ electrons}$$

Q: 2(iv) _____
A: 2(iv) _____

Electron affinity:-

the amount of energy released when an electron is added to a chemical neutral, gaseous atom to form a negative ion.

Across a period:- It increases - atomic size decreases and nuclear charge increases.

Down a group:- It decreases - atomic size increases and nuclear charge decreases.

Q: 2(v) (or) _____
A: 2(v) (or) _____

Mg (2, 8, 2) has 2 valence electrons
O (2, 6) needs 2 electrons to complete its octet.

Mg loses 2 electron (Mg^{2+} , 2, 8 neon E.C.)
O has 2 electron now $\rightarrow$ (O^{2-} , 2, 8) (neon E.C.)
The oppositely charged Mg^{2+} and O^{2-} ions attract by a strong electrostatic force = ionic bond, forming MgO.

Q. 2(vi)
A. 2(vi)

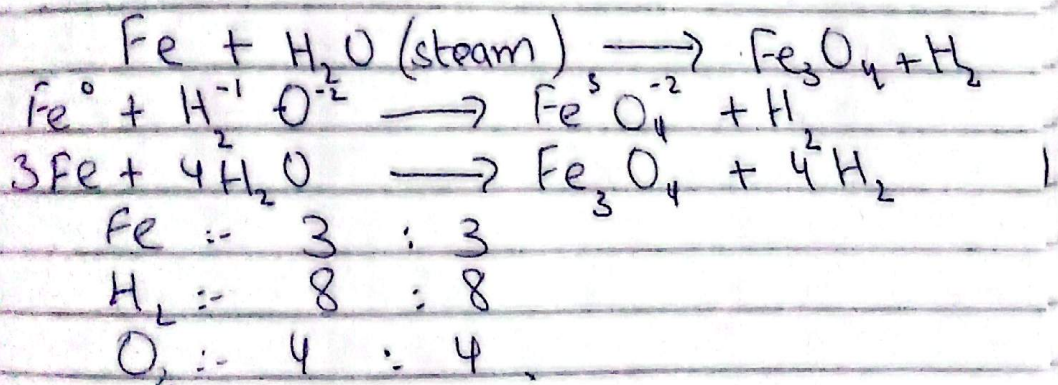
Molar mass :-

mass in grams of 1 mole of a substance, it is equal to the formula / molecular mass (in amu) it is expressed in g/mol.

~~Na₂CO₃~~
Na₂CO₃ :-

$$\begin{aligned} & 2(23) + 12 + 3(16) \\ & 46 + 12 + 48 \\ & = 106 \text{ g/mol.} \end{aligned}$$

Q. 2(vii) (or)
A. 2(vii) (or)



Q: 2 (viii)

A: 2 (viii)

(a) Mn in MnO_2

$$x + 2(-2) = 0$$

$$x - 4 = 0$$

$$x = +4$$

(b) Cl in HClO_4 :-

$$1 + x + 4(-2) = 0$$

$$1 + x - 8 = 0$$

$$x - 8 + 1 = 0$$

$$x - 7 = 0$$

$$x = +7$$

(c) P in H_3PO_4

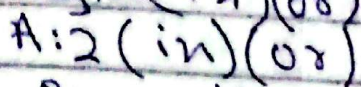
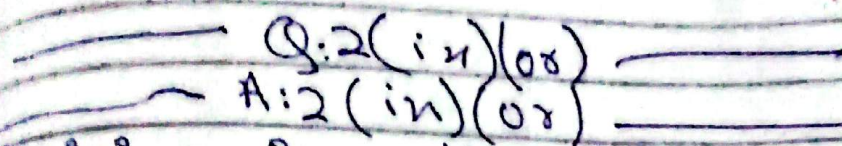
$$3(1) + x + 4(-2) = 0$$

$$3 + x - 8 = 0$$

$$x - 8 + 3 = 0$$

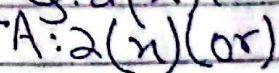
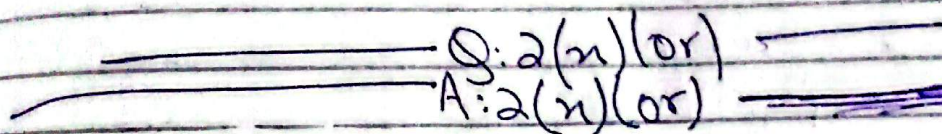
$$x - 5 = 0$$

$$x = +5$$



Aluminium is reactive and reacts quickly with oxygen air but it forms a thin, hard, continuous layer of Aluminium oxide (Al_2O_3) on its surface.

This oxide layer is impervious and sticks tightly, sealing the metal underneath so that no further air or moisture can reach it; metal below is therefore protected from further corrosion.



Catalyst:-

A substance that increases the rate of a chemical reaction without itself being consumed (it is chemically unchanged at the end).

Industrial example:-

Fe in the Haber process ($\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$)

① It increases the rate by providing an alternative path of lower activation energy.

② It does not change ΔH , the overall enthalpy change stays the same.

Q: 2 (i) (a)
A: 2 (i) (a)

Physical equilibrium

① Balance in a reversible physical change & no new substance is formed.

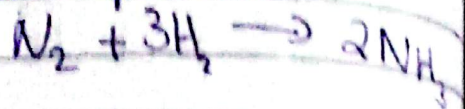
Example :-

~~water~~
water $\rightleftharpoons$ water vapour
in a closed bottle.

Chemical equilibrium

Balance in a reversible chemical change & new substance is formed.

Example :-



Q: 3 (i)

A: 3 (i)

Electronic Configuration:-

the arrangement of the electron in of the atom in its shells (and sub-shells) around the nucleus.

① Carbon (Z=6) :: $1s^2, 2s^2, 2p^2$

group :- $2+2 = 4 + 10 = 14$

IV or 14

period :- 2nd

② Sulphur (Z=16) :: $1s^2, 2s^2, 2p^6, 3s^2, 3p^4$

group :- $4+2 = 6 + 10 = 16$

VI or 16

period :- 3rd

Q An atom with 12 protons and 12 electrons is Mg (magnesium),
EC (Electronic Configuration) ?
 $1s^2, 2s^2, 2p^6, 3s^2$.

Q It has is a metal, bcz it has only 2 electron in its valence shell, which it tends to lose (metal are electron donors).

Q It loses 2 electrons to form the ion Mg^{2+} (charge: +2).

Q Its valency :- ii or 2.

Q : 3(ii)

A : 3(ii)

Q Empirical formula :-

is the formula that gives simplest whole-number ratio of atoms of each elements in a ~~compound~~ compound.

Q Molecular formula :-

is the formula that gives the actual number of atoms of each element, in a compound or one molecule.

Empirical formula mass of ~~C₆H₁₂O~~
~~C₆~~ CH₂O :-

$$12 + 1(2) + 16 = 12 + 2 + 16 \\ 14 + 16 \\ 30$$

$n = \frac{\text{molecular mass}}{\text{empirical formula}}$

$$\frac{180}{30} = 6$$

molecular formula :-
(CH₂O)₆

= C₆H₁₂O₆ (glucose).

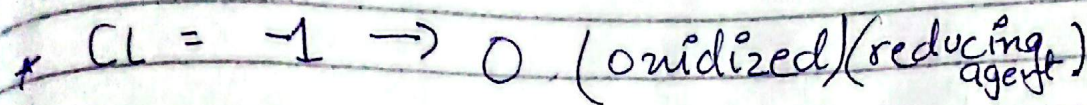
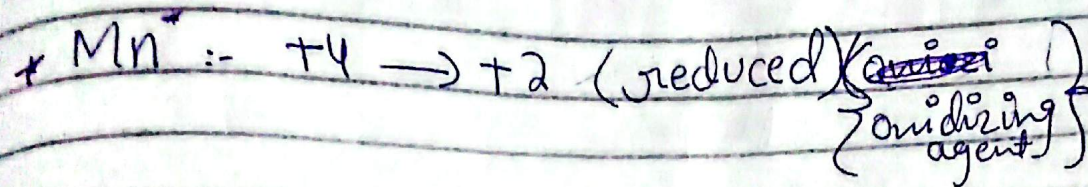
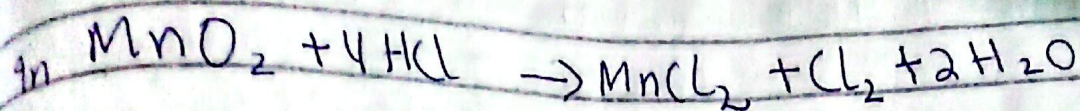
Q: 3 (iii) (or) _____
A: 3 (iii) (or) _____

Oxidizing agents:-

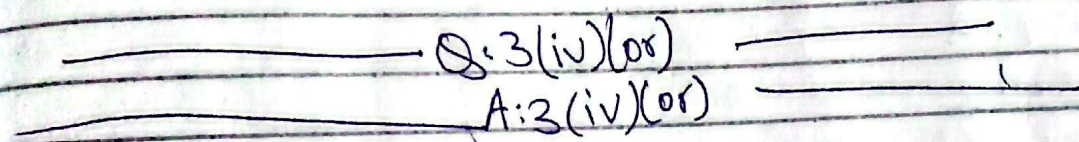
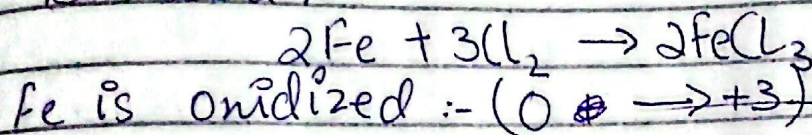
The substance that causes another substance to gain electrons and is itself reduced.

Reducing agents:-

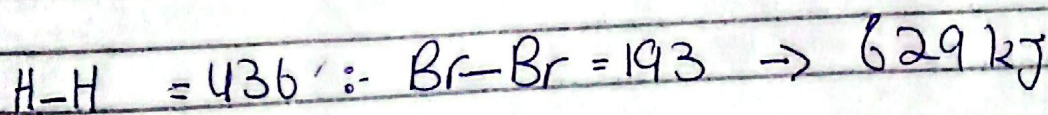
Substance that causes another substance to lose electrons and is itself oxidized.



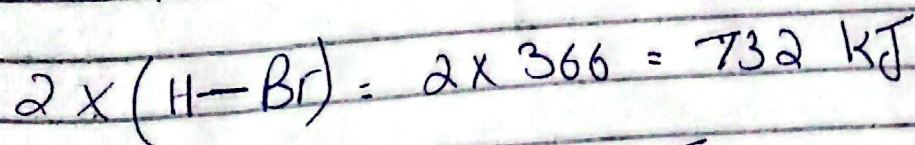
⑥ Balance d :-



Bonds broken (reactants) :-



Bonds formed (products) :-



$$\Delta H = 629 - 732 = -103 \text{ J}$$

ΔH is negative :- This reaction is exothermic

⑦ (i) Respiration :- exothermic, ΔH is negative (energy is released)

Photosynthesis :- endothermic, ΔH is positive
(energy from sunlight is absorbed).

MCQs

1. A
2. B
3. C
4. D
5. C
6. D
7. A
8. B
9. D
10. A
11. B
12. D